

High Level Tasks

0. Getting the Foundation ready

- GitHub Repo & add everyone as collaborators
- First PR : CLAUDE.md, SPEC.md
- Decide on Tentative deadlines (Fail fast)
- Repo Structure - can be decided as and when we raise PRs

1. Datasets

a. Custom dataset generation:

- i. Drift - Sowmya, Harish
- ii. Changepoints (easier) - Goutham, Meena, Dinesh
- iii. Outlier Flagging - Nadhiya, Amudhaval

1. Look up: Darts Timeseries Generation

Link: [© Utils for TimeSeries generation — darts documentation](#)

- 2. Have Knobs/parameters to adjust start time, drift/changepoints parameters, number of outliers etc

b. Standard Datasets with the same:

- Drift
- Changepoints (easier)
- Outlier Flagging

2. Problem Reproduction

- a. Use datasets from above and run Prophet, ARIMA and Seasonal Naive model (again use Darts), reproduce the issue
- b. Construct the train-test set (filter out datasets where we don't see the issue)

3. Agent Development (Diagnosis → Intervention → Backtest → Recommendation)

- a. A simple ReAct Agent with tools (input to the agent is a matplotlib graph of historical data and forecast)

i. Tool development with unit tests:

1. Drift Identification:

- 2. Changepoint Identification: [Time Traveling with Data Science: Focusing on Change Point Detection in Time Series Analysis \(Part 2\) - Blog des Fraunhofer IESE](#)

3. Backtest

- a. Integration with Agent and verify that the agent is able to use tools and diagnose issues from the train dataset.

(Prompts have to be versioned)

4. Report Generation

4. **Agent Evals**

a. Manual Evaluation, Failure mode identification

b. LLM Judge deployment

i. **Versioned Prompts**

ii. LLM Judge Evaluation

5. (Optional) **Chat interface / UI :**

Input is graph and timeseries data ; Output is the report and again optionally a dynamic widget

Reading/Learning:

1. TimeSeries Forecasting Text Book: [Forecasting: Principles and Practice \(3rd ed\)](#)
2. Introduction to LangGraph: <https://academy.langchain.com/courses/intro-to-langgraph>
3. (Optional, I will be showing you these anyways):
<https://academy.langchain.com/courses/intro-to-langsmith>

Cool Stuff:

DeepSeek Paper: Thinking with Visual Primitives

Video: [DeepSeek's New AI Is A Game Changer](#)

Paper: [GitHub - ailuntx/Thinking-with-Visual-Primitives: Archived snapshot of Thinking-with-Visual-Primitives](#)

Paper: [Thinking_with_Visual_Primitives.pdf · NodeLinker/deepseek-ai-Thinking-with-Visual-Primitives-deleted-repo at main](#)